

AMD Developer Hackathon: ACT II · Unicorn Track



Crop Advisor Lite

AI crop advisory for Pakistan's 8 million smallholder farmers – in English and Urdu.

Gemma 3 27B · Fireworks AI (AMD-hosted inference) · Streamlit · Docker

The problem

- **8M+** smallholder farms in Pakistan (1-12 acres) drive the country's food supply.
- Most farmers have **no affordable access to agronomic advice**: what to sow after harvest, when the sowing window closes, what an acre costs in inputs.
- Government extension services are stretched thin; advice travels by word of mouth – often late, often wrong.
- Bad crop decisions compound: one wrong season can wipe out a smallholder's yearly margin.

The solution – a chat advisor in the farmer's own language

I have 5 acres in Multan, just harvested wheat. Water is short. What should I sow next?

Crop Advisor Lite: With short water after wheat in Multan, consider **mungbean** (60–90 day crop, low water, fixes nitrogen) or **sesame** before the cotton window. Avoid rice this kharif – it needs far more water than a tubewell can affordably supply. Rough input cost for mungbean: PKR 25,000–40,000/acre...

- Decision-ready answers: named crops, months, rupee ranges – not vague generalities.
- English in, English out. Urdu in, Urdu out.
- Honest about limits: refuses to invent market prices; routes high-stakes calls to extension offices.

Built on AMD

- Model** **Gemma 3 27B Instruct** – Google DeepMind’s open-weight model: multilingual, lightweight, ideal for low-cost farmer-facing advice
- Inference** **Fireworks AI API** – AMD-hosted inference for this hackathon; serverless, pay-per-token
- App** Streamlit (Python) – chat UI with example prompts in English and Urdu
- Packaging** Docker – one command to run anywhere: `docker run -p 8501:8501 -e FIREWORKS_API_KEY=...`

What's next

- **Grounding:** retrieval over provincial extension guides and crop calendars, so every answer cites a real source.
- **Urdu voice:** speech in/out for farmers who can't read or type comfortably.
- **WhatsApp delivery:** meet farmers on the platform they already use daily.
- **Cost path:** Gemma's small footprint on AMD inference keeps per-farmer cost near zero – the unit economics of free advice.